

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study ¹
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Submission Date	30/08/26

1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalisation.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2 Problem Statement

Given 30 numerical features describing a breast cell nucleus, the task is to classify the sample as Benign or Malignant. A single Decision Tree is trained first, then extended into a Random Forest ensemble, with hyperparameters for both selected using 5-fold cross-validation. The goal is to see how much the ensemble improves generalisation over the single tree.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30 numerical features
Number of Classes	2 (Benign, Malignant)
Missing Values	None
Train-Test Split	80% – 20%, stratified (455 train, 114 test)

4 Exploratory Data Analysis

Class Distribution

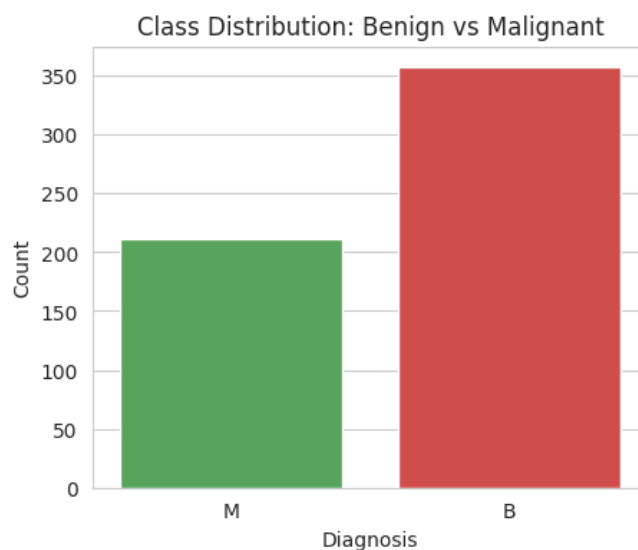


Figure 1: Class distribution: Benign vs Malignant

Inference: Out of 569 samples, 357 are Benign and 212 are Malignant, roughly a 63:37 split. The dataset is mildly imbalanced but not to a worrying degree, so accuracy alone is not fully relied on and F1/recall on the malignant class is tracked more closely, since missing an actual malignant case matters more than a false alarm. Stratified splitting keeps this ratio consistent in the train and test sets.

Correlation Heatmap

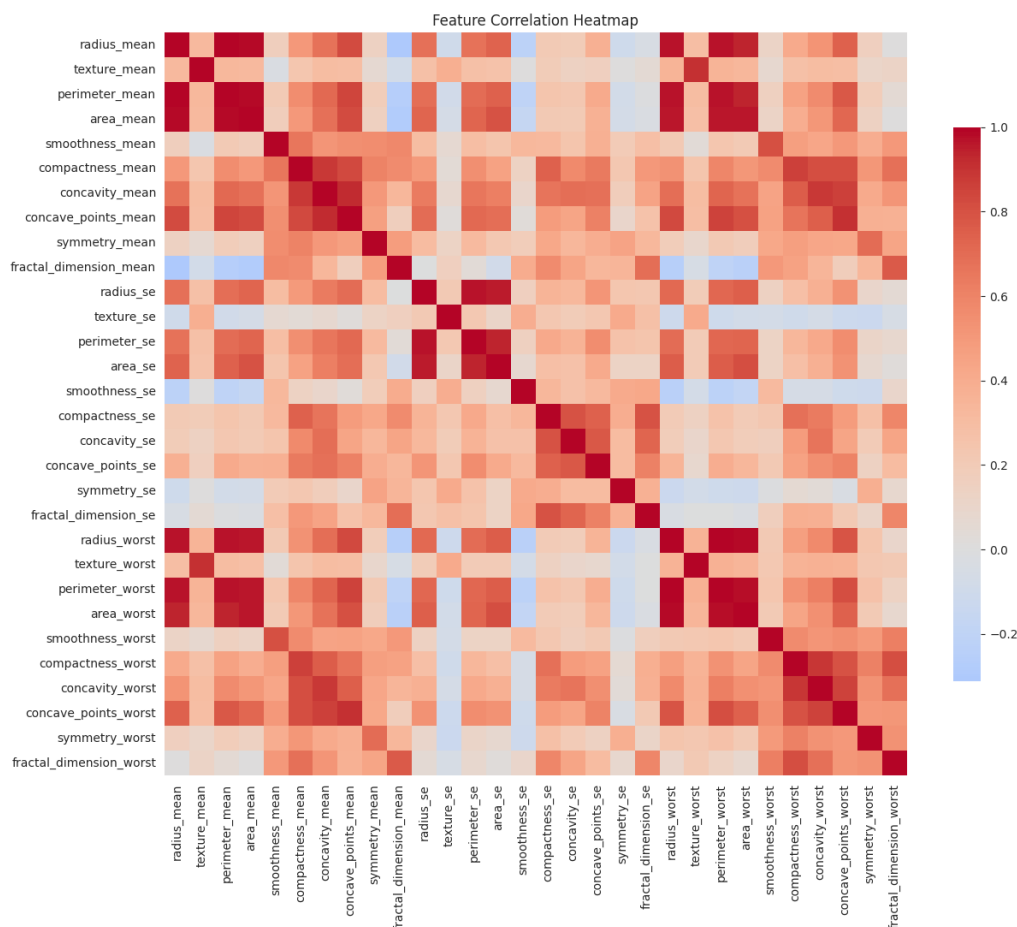


Figure 2: Feature correlation heatmap

Inference: Many of the mean, standard-error and worst versions of the same measurement (radius, perimeter, area) are strongly correlated with each other, since they come from the same underlying cell shape. This kind of overlap would hurt a linear model but barely affects tree-based models like Decision Tree and Random Forest, since a tree only ever splits on one feature at a time.

Feature Correlation with Diagnosis

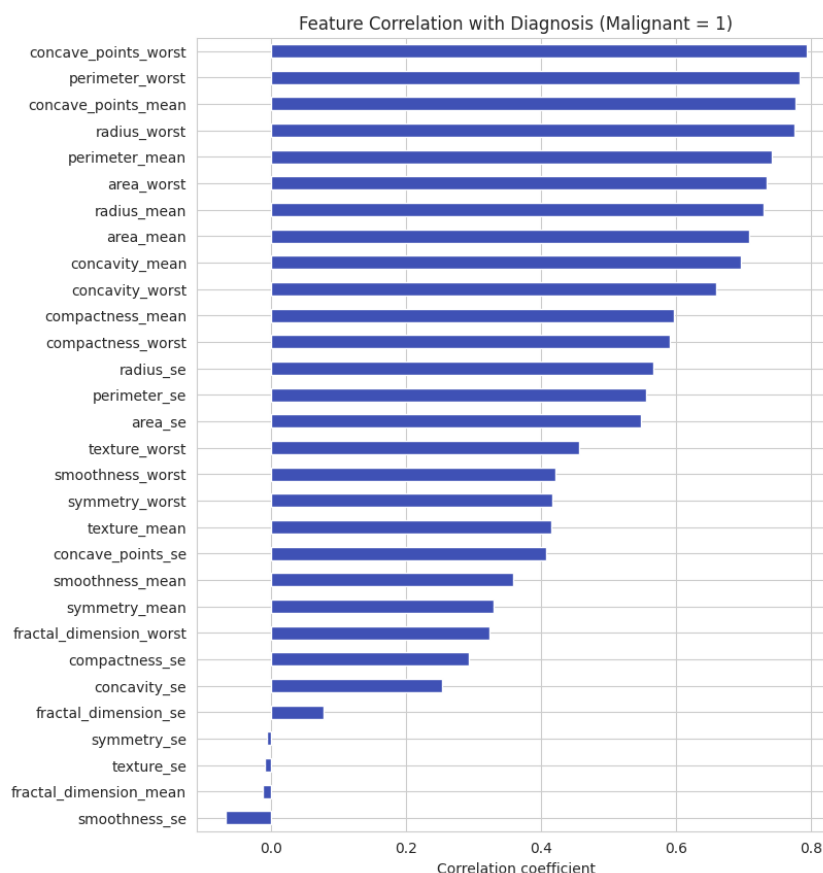


Figure 3: Feature correlation with diagnosis (Malignant = 1)

Inference: Concave points, perimeter, radius and area (mostly the worst variants) show the strongest positive correlation with malignancy. This matches the general medical intuition that larger, more irregular cell nuclei are more likely to be malignant, and these same features are expected to dominate the splits chosen by both tree models.

5 Data Preprocessing

- **Missing values:** None found, so no imputation was required.
- **Encoding:** Diagnosis label mapped to 1 = Malignant, 0 = Benign.
- **Feature scaling:** Not required, since Decision Tree and Random Forest split on raw feature thresholds and are not distance-based.
- **Train-test split:** 80–20 stratified split, giving 455 training and 114 testing samples, keeping the roughly 63:37 class ratio the same in both.

6 Machine Learning Algorithms

Decision Tree

A Decision Tree recursively splits the feature space using an impurity measure (Gini index or entropy) to form simple decision rules. **Advantage:** easy to interpret and visualise. **Limitation:** prone to high variance – an unrestricted tree can grow very deep and memorise the training data instead of generalising.

Random Forest

Random Forest is an ensemble of many Decision Trees, each trained on a different bootstrap sample of the data and considering only a random subset of features at each split. The final prediction is decided by majority voting across all trees. **Advantage:** averaging over many de-correlated trees reduces variance and improves generalisation compared to a single tree. **Limitation:** less interpretable than a single tree, and slower to train and predict.

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	1.x
IDE	Jupyter Notebook
Operating System	Linux
Hardware	Standard CPU (no GPU required)

8 Hyperparameter Selection

Hyperparameters were chosen using `GridSearchCV` with 5-fold stratified cross-validation, scored on F1 rather than plain accuracy since correctly identifying malignant cases matters [more here](#).

Model	Best Parameters
Decision Tree	<code>criterion=entropy, max_depth=4, min_samples_split=2, min_samples_leaf=2</code>
Random Forest	<code>n_estimators=50, max_depth=10, max_features=sqrt, bootstrap=False</code>

9 Results

Metric	Decision Tree	Random Forest
Accuracy	0.9298	0.9649
Precision	1.0000	1.0000
Recall	0.8095	0.9048
F1-score	0.8947	0.9500
ROC-AUC	0.956	0.995

10 Result Visualization

Decision Tree Structure

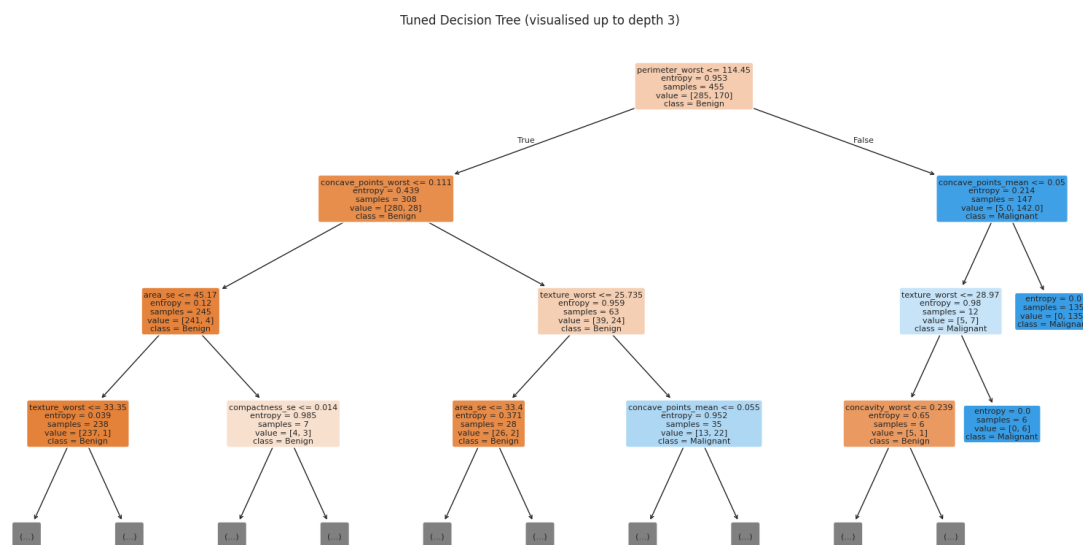


Figure 4: Tuned Decision Tree, visualised up to depth 3

Inference: Only the top few levels of the tuned tree are shown here for readability. The root split uses `perimeter_worst`, followed by `concave_points_worst` and `concave_points_mean` – exactly the features flagged as most correlated with malignancy earlier, which confirms the tree is picking up genuinely informative splits rather than noise.

Random Forest Feature Importance

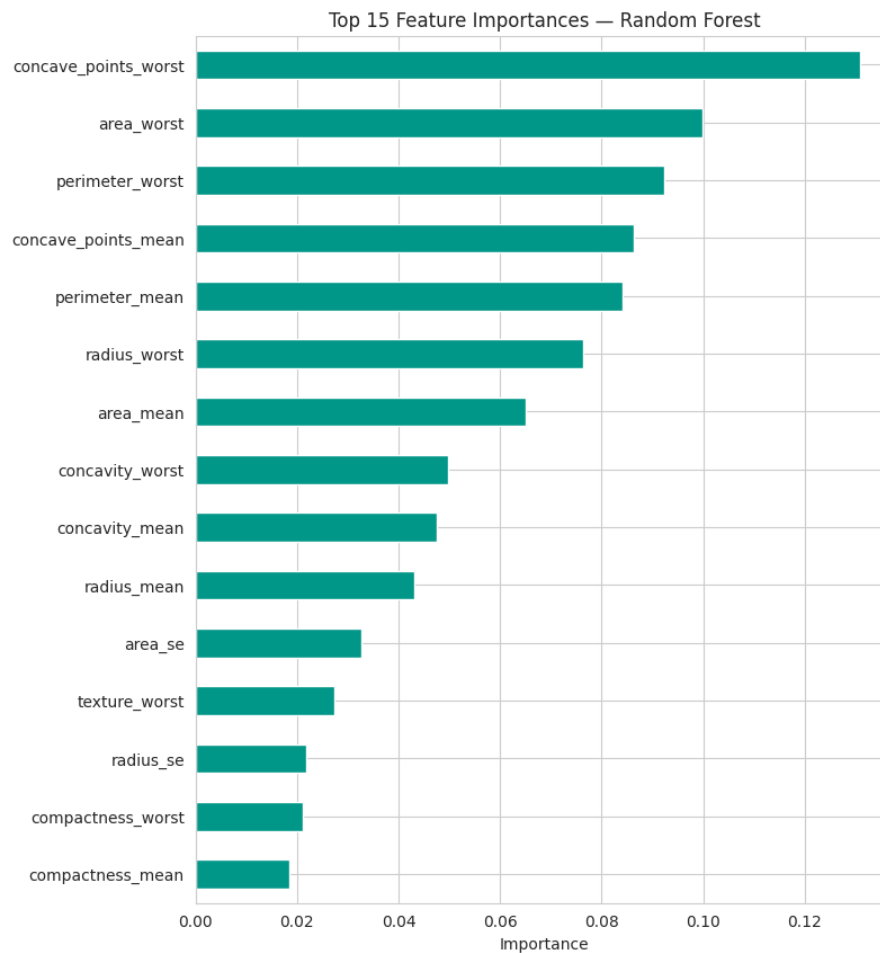


Figure 5: Top 15 feature importances from the Random Forest

Inference: `concave_points_worst`, `area_worst` and `perimeter_worst` come out as the most important features for the Random Forest, agreeing closely with the correlation analysis done during EDA. This agreement between two independent methods adds confidence that these features genuinely drive the classification decision.

Confusion Matrices

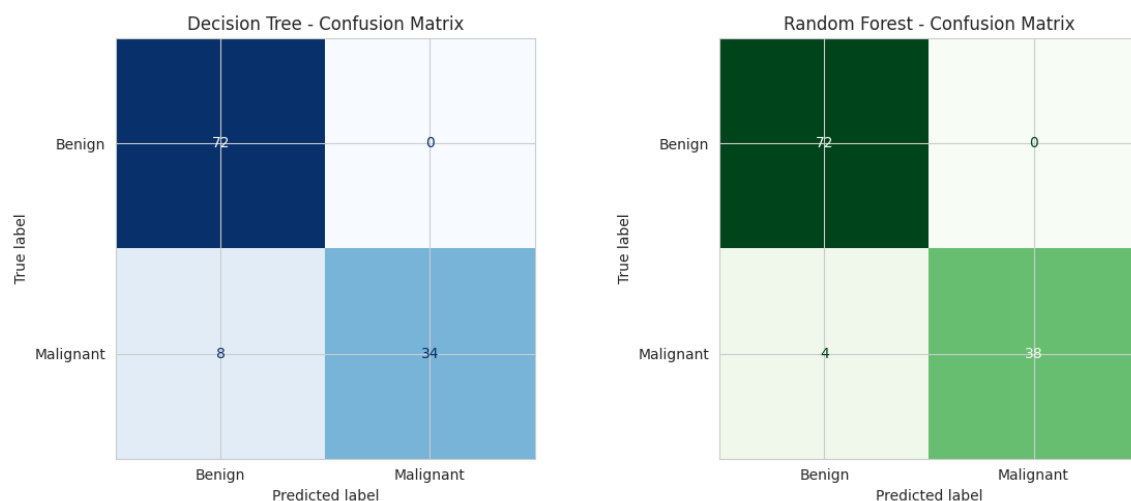


Figure 6: Confusion matrices: Decision Tree vs Random Forest

Inference: The Decision Tree misses 8 malignant cases (false negatives) while the Random Forest misses only 4, cutting the missed-malignant count in half. Since a missed malignant case is the costliest kind of mistake in this setting, this is the clearest sign of the Random Forest's improvement over the single tree.

ROC Curve

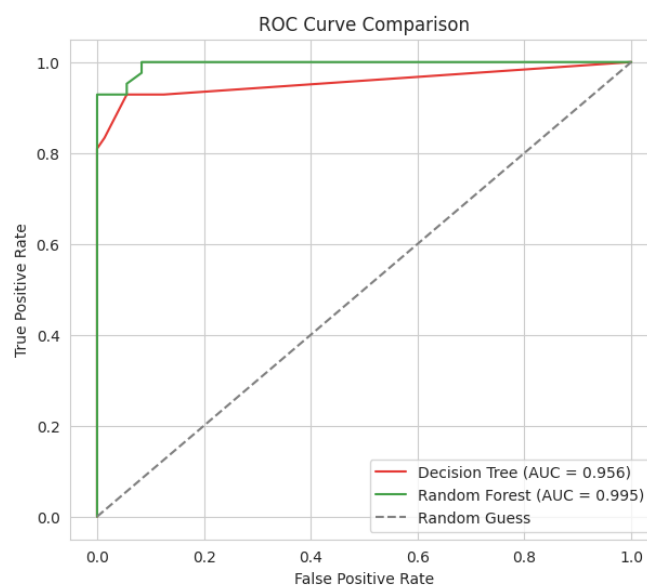


Figure 7: ROC curve comparison

Inference: The Random Forest curve hugs the top-left corner much more closely than the Decision Tree curve, and its AUC (0.995) is clearly higher than the Decision Tree's (0.956). This shows the Random Forest separates the two classes better across all thresholds, not just the default 0.5 cutoff used for the confusion matrix above.

Metric Comparison

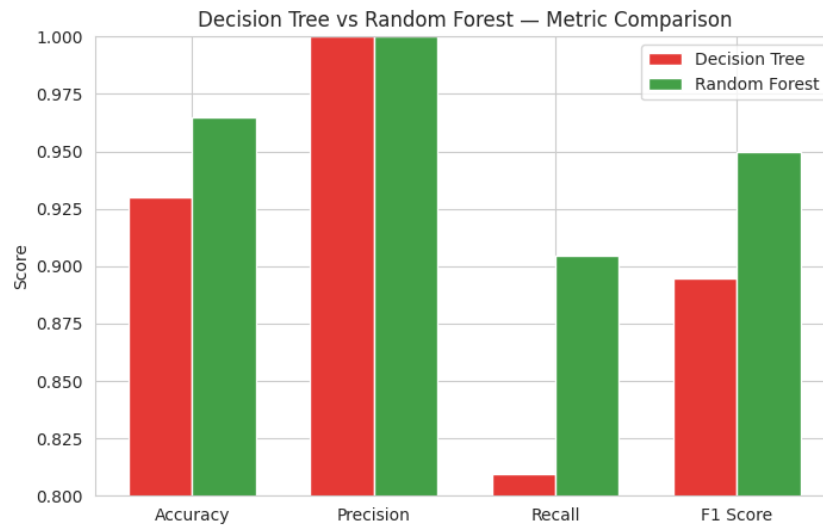


Figure 8: Decision Tree vs Random Forest – metric comparison

Inference: The Random Forest matches the Decision Tree on precision (both 1.0, meaning neither raises false alarms) but is clearly ahead on accuracy, recall and F1-score. Recall shows the biggest gap (0.81 vs 0.90), which lines up with the confusion matrix observation that the Random Forest misses fewer malignant cases.

11 Discussion

The baseline unrestricted Decision Tree overfits, growing to a depth of 8 with 24 leaves on the training data. Once tuned with cross-validation, the tree is restricted to depth 4, trading a bit of training fit for better generalisation, but it still misses a meaningful fraction of malignant cases on the test set (recall 0.81). The Random Forest, by averaging 50 de-correlated trees, pushes recall up to 0.90 and F1 up to 0.95 without any loss in precision. The 5-fold cross-validation fold scores also show the Random Forest to be more stable (average CV accuracy 0.971 vs 0.941 for the Decision Tree), confirming that ensembling reduces variance rather than just getting lucky on one train-test split. One limitation is that this dataset is fairly clean and well-separated, so even the single Decision Tree performs reasonably well; on a noisier or larger dataset the gap between the two models could be more pronounced.

12 Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation on the Wisconsin Diagnostic Breast Cancer dataset. Hyperparameters were selected based on average cross-validation performance to ensure robust generalisation. The tables below summarise the cross-validation tuning results and the fold-wise comparison, as required by the experiment.

Table 1: Decision Tree Hyperparameter Evaluation (Top results, 5-Fold CV)

Criterion	Max Depth	Min Samples Split	Min Samples Leaf	Avg CV F1
entropy	4	5	2	0.9163 (Best)
entropy	4	2	2	0.9163
entropy	4	2	1	0.9145
entropy	4	10	2	0.9142
entropy	5	10	2	0.9121

Table 2: Random Forest Hyperparameter Evaluation (Top results, 5-Fold CV)

n_estimators	Max Depth	Max Features	Bootstrap	Avg CV F1
50	10	sqrt	False	0.9617 (Best)
200	10	sqrt	False	0.9615
200	None	sqrt	False	0.9615
200	5	sqrt	False	0.9608
50	None	sqrt	False	0.9589

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9341	0.956	0.9011	0.956	0.956	0.9407
Random Forest	0.9670	0.989	0.9670	0.956	0.978	0.9714

Table 4: Final Test Set Performance Comparison

Metric	Decision Tree	Random Forest
Accuracy	0.9298	0.9649
Precision	1.0000	1.0000
Recall	0.8095	0.9048
F1 Score	0.8947	0.9500

Overall, the Random Forest consistently outperforms the single Decision Tree on every metric except precision, where both are tied, and it is also more stable across cross-validation folds. This confirms the theoretical expectation that bagging and random feature selection reduce variance and improve generalisation over a single decision tree.

13 References

1. scikit-learn developers, “Decision Trees,” scikit-learn documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/tree.html>
2. scikit-learn developers, “Random Forests,” scikit-learn documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/ensemble.html#forest>
3. W. N. Street, W. H. Wolberg, and O. L. Mangasarian, “Nuclear feature extraction for breast tumor diagnosis,” *IS&T/SPIE 1993 International Symposium on Electronic Imaging*, 1993.
4. UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic) Data Set.” [Online]. Available: <https://archive.ics.uci.edu>